

1 Project Data Simulation: Usability of Artificial Intelligence in Coaching for Physical Activity Promotion (AlcoPA)

Documentation

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Abbreviations

AlcoPA	<i>Usability of Artificial Intelligence in Coaching for Physical Activity Promotion</i>
BMZI	<i>Bernese motive and goal inventory</i>
CG	<i>Control Group</i>
IBCM	<i>Integrated Behavior Change Model</i>
IG	<i>Intervention Group</i>
KIM	<i>Kompetenz-Intentions-Modell</i>
MET	<i>Metabolic equivalent of task</i>
PA	<i>Physical Activity</i>
PBC	<i>Perceived Behavioral Control</i>
SD	<i>Standard Deviation</i>
SDT	<i>Self-Determination Theory</i>
SRBAI	<i>Self-Report Behavioral Automaticity Index</i>
T1	<i>Timepoint 1</i>
T2	<i>Timepoint 2</i>
T3	<i>Timepoint 3</i>
TAM	<i>Technology Acceptance Model</i>
TPB	<i>Theory of Planned Behavior</i>

1 Purpose of the Simulation

The present simulation was developed to generate a realistic synthetic dataset for the physical activity intervention study within the Usability of Artificial Intelligence in Coaching for Physical Activity Promotion (AlcoPA) project. The primary objective was to create data that resemble empirically observed patterns in health behavior research while maintaining full control over the underlying data-generating process.

The simulated dataset is primarily intended to enable the testing of the complete study workflow across multiple survey time points. This includes the validation of survey logic, such as filtering mechanisms and data importation from different time-points. In addition, the dataset supports the evaluation of the full data pipeline, from raw survey export to processed datasets ready for analysis. This includes testing data transformation procedures and ensuring that all variables are correctly structured and interpretable.

While the simulated data can also be used to test statistical analysis approaches (e.g., linear mixed models or structural equation models), this represents a secondary objective. Further, it is possible to investigate the validity of the simulated data against the data gathered by the real-world measurement timepoints (i.e., T1-3) of the AlcoPA project. The focus here lies though on ensuring that the end-to-end workflow from data generation to final dataset is robust and reproducible.

2 Overview of the Workflow

The simulation pipeline consists of three main components, implemented as separate R scripts that can be executed in an R environment (e.g., via RStudio, VS Code, or command line).

1. **Data simulation script:** This script generates the core dataset, including participant-level characteristics, longitudinal measurements, psychological constructs, physical activity outcomes, and item-level survey responses.
2. **Mapping and export script:** The simulated data are transformed into a format that closely mirrors a real LimeSurvey export. This includes variable renaming, recoding into categorical labels, generation of technical variables, and alignment with an existing survey template.
3. **Simulation check script:** This script allows to evaluate the final dataset with respect to structural properties, intervention effects, distributional characteristics, and theoretical plausibility. This step supports iterative adjustment of simulation parameters in order to create a realistic synthetic dataset.

The workflow produces the following key outputs:

- A simulation dataset in CSV format containing all generated variables
- A template-compatible Excel file structured like a LimeSurvey export
- Diagnostic console outputs used to evaluate the realism and consistency of the simulated data

Overall, the pipeline separates data generation, formatting, and validation, ensuring reproducibility at each stage.

3 Theoretical and Empirical Foundations

3.1 Evidence-Based Parameter Selection

The simulation is informed by the current literature focusing on parameters relevant to constructs of interest in AlcoPA (see Appendix A1). Two complementary sources of evidence were used:

Meta-analyses and systematic reviews, primarily providing correlations, standardized effect sizes, and theoretical relationships between constructs (e.g., intention–behavior associations in the Theory of Planned Behavior).

Validation and empirical studies, providing raw score distributions (means and standard deviations) and reliability estimates for established measurement instruments (e.g., motives and goals, habit automaticity, planning scale).

For several constructs, particularly within the Theory of Planned Behavior and the Technology Acceptance Model, no consistent pooled baseline means and standard deviations are available due to heterogeneity in measurement scales across studies. In these cases, plausible parameter ranges were defined based on theoretical expectations and typical value ranges reported in the literature.

Consequently, the simulation combines empirically grounded parameters (e.g., Bernese motive and goal inventory (BMZI) means, planning scale distributions), and theory-informed and pragmatically calibrated parameters (e.g., Theory of Planned Behavior (TPB) and Technology Acceptance Model (TAM) baseline values, intervention effect sizes). This hybrid approach ensures both realism and flexibility in the generated data.

3.2 Modeled Psychological Frameworks

The simulation integrates multiple theoretical frameworks used in physical activity and health behavior research:

Habit Theory (Gardner & Lally, 2018): Habit strength is modeled as a key determinant of behavior, informed by research on automaticity and habit formation.

Integrated Behavior Change Model (IBCM; Hagger & Chatzisarantis, 2014): Core constructs include intention, attitude, injunctive and descriptive norms, and perceived behavioral control. Relationships between these constructs and behavior follow meta-analytically supported patterns. Intrinsic and extrinsic motivation, as well as perceived competence and perceived freedom of choice, were operationalized as scale means based on established self-determination theory (SDT) measures (range: 0–4). The simulation reflects typical empirical patterns, with higher intrinsic than extrinsic motivation among participants.

Motivational Control, Volitional Self Control and Planning (Welcher Sport für wen?; Schmid et al., 2024): Action planning and self-control are included as key post-intentional processes influencing behavior.

Motivational constructs (BMZI; Lehnert et al., 2011; Gut et al., 2019): Domain-specific motives for physical activity are modeled using empirically informed distributions, particularly for BMZI subscales.

Technology Acceptance Model (TAM; Davis, 1985; Gómez-Ruiz et al., 2022): Constructs such as perceived usefulness, ease of use, enjoyment, trust, and intention to use are included to model engagement with the intervention technology. Their structural relationships are based on established TAM frameworks.

Together, these frameworks form a comprehensive representation of psychological and behavioral processes underlying physical activity.

3.3 Sociodemographic Effects

In addition to psychological and behavioral constructs, the simulation incorporates key sociodemographic variables, including for instance age, gender, income, and prior app usage.

These variables are modeled as exogenous predictors influencing both: physical activity outcomes, and app usage behavior. The direction of these effects is informed by empirical literature on physical activity and digital health interventions (Pontin et al., 2021). For example: women, participants aged 25-50 years, and participants with prior app experience are assigned a higher probability of app use, whereas high income reduces app-use. For physical activity, age has a small negative effect, while male gender, prior app use, and high income have positive effects on daily steps and weekly metabolic equivalent of task (MET)-minutes.

While the inclusion and direction of these effects are evidence-based, the magnitude of the effects was specified to reflect small effect sizes typically observed in digital physical activity interventions (Singh, 2024).

4 Script 1: Data Simulation

4.1 Simulated Study Design

The simulation is based on a longitudinal study design with three measurement time points (timepoint 1 (T1), timepoint 2 (T2), timepoint 3 (T3)) and a total sample size of $N = 200$ participants. Each participant is assigned to one of two study groups: Intervention Group (IG) or Wait-List Control Group (CG)

The design follows a cross-over structure:

- At T1, the intervention is administered to the IG, while the CG remains untreated.
- At T2, the CG receives the intervention, whereas the IG is no longer actively treated.

In addition, a carryover effect is implemented for the IG at T3, reflecting residual effects of the intervention after its active phase. This design allows for the simulation of time-dependent treatment effects and group-by-time interactions.

4.2 Person-Level Latent Structure

To represent stable individual differences across participants, a set of latent person-level variables is generated using a multivariate normal distribution. These variables represent baseline tendencies for key psychological constructs (e.g., habit, intention, and autonomous motivation, which is later operationalized through intrinsic motivation, perceived competence, and perceived choice).

A predefined correlation matrix is used to induce realistic associations between constructs, based on meta-analytic evidence (e.g., correlations among TPB variables and between autonomous motivation and behavior). These latent variables function as random intercepts, ensuring that individuals differ systematically in their baseline levels, and correlations between constructs are preserved across all time points.

4.3 Intervention and Time Effects

Time-varying treatment effects are implemented using indicator variables: A treatment indicator specifies whether a participant is currently exposed to the intervention. A carryover indicator models residual effects in subsequent time points.

For each construct, values are generated as a function of:

- a baseline mean,
- the person-specific latent deviation,
- treatment and carryover effects,
- and random within-person noise.

4.4 Generation of Latent Constructs

A range of psychological constructs is simulated as latent variables at each time point. These include:

- Habit (automaticity)
- Intention, attitude, injunctive norms, descriptive norms, and perceived behavioral control
- Autonomous motivation (operationalized through intrinsic motivation, perceived competence, and perceived choice)
- Action planning, motivational competence, and self-control
- Additional motivational constructs (e.g., motive and goal related variables)

Each latent construct is generated according to a general model:

$$\text{latent value} = \text{baseline mean} + \text{person-level effect} + \text{treatment effect} + \text{carryover effect} + \text{random noise}$$

Baseline means and variances are informed by empirical studies where available (e.g., BMZI, planning scales), and otherwise calibrated within plausible ranges based on theory.

4.5 Generation of Physical Activity Outcomes

Physical activity is represented using weekly MET-minutes as the primary outcome. Baseline activity levels are defined to reflect a low-active population profile, based on a simple composition of light, moderate, and vigorous activity, resulting in an initial weekly MET value of approximately 410.

Intervention and follow-up effects are informed by step-based empirical findings. Reported changes in daily step counts are translated into weekly MET-minutes using a constant conversion factor, assuming 100 steps per minute at an intensity of 3 MET across seven days. These converted values are incorporated as treatment and carryover effects within the MET-based model.

Weekly MET-minutes are generated as a function of:

- a baseline intercept
- a participant-specific random intercept capturing stable individual differences
- treatment and carryover effects
- sociodemographic influences (see Section 3.3)
- a behavioral driver combining key psychological predictors (habit, intention, perceived behavioral control, and intrinsic motivation)
- random residual variation.

4.6 Simulation of Survey-Based PA Measures

The survey-style PA variables are derived from the simulated weekly MET-minutes. The script allocates each participant's total MET-minutes to light (92 %) and moderate (8%) activity shares. Light activity remains dominant, moderate activity becomes somewhat larger in treatment or carryover phases, and vigorous activity is very small and stays mostly at zero because of the low total activity level of the population.

After the share allocation, weekly minutes are computed separately for light, moderate, and vigorous activity by dividing MET-minutes by the corresponding intensity weights (3, 5, and 9 MET; (Nigg et al., 2024)). Days per week are then derived by dividing weekly minutes by target daily values (35 minutes for light, 25 for moderate), rounding, and truncating to the range 0-7.

4.7 Item Generation

Observed items are generated by adding measurement error to the latent construct scores and then transforming the resulting values into the response range of each scale. All items are generated on ordinal Likert scales. The following ranges are used: 1–5 (habit, self-control, motivational competence, BMZI), 1–7 (TPB variables, TAM), 0–4 (intrinsic motivation, perceived competence, perceived choice, extrinsic motivation), and 1–4 (action planning).

Because items are generated from shared latent variables with item-specific noise, each scale shows within-construct consistency while still preserving realistic response variability.

4.8 Technology Acceptance Model

TAM variables are simulated only at time points where participants are exposed to the intervention with the artificial intelligence coaching app so for the IGat T2 and for the CG at T3

Latent TAM factors are generated and used to derive the observed items, including perceived usefulness, ease of use, enjoyment, trust and intention to use. In addition, app usage behavior is modeled probabilistically using a logistic function, with predictors including:

- sociodemographic variables (age, gender, prior app use),
- and treatment exposure.

4.9 Missing Data Mechanisms

Multiple types of missing data are simulated to reflect realistic study conditions:

- **Monotone dropout:** A subset of participants drops out after T1 or T2, resulting in missing data at later time points.
- **Random missingness:** A small proportion of item-level responses is set to missing completely at random.
- **Structural missingness:** Certain variables (e.g., TAM items) are only observed under specific conditions (e.g., intervention exposure).

4.10 Output

The simulation script produces a dataset in CSV format containing:

- participant identifiers and study design variables
- observed items of latent psychological constructs
- physical activity items and outcomes (i.e., MET-minutes)
- sociodemographic variables

This dataset serves as the input for the subsequent mapping and export step.

5 Script 2: Mapping to Survey-Compatible Format

5.1 Purpose and General Approach

The mapping script converts the simulation output into an Excel file that resembles a real LimeSurvey response export. It first reads a genuine survey export template and creates an Excel file output with identical column names. Simulated variables are then inserted, recoded, or derived so that the result is structurally compatible with the survey workflow.

5.2 Direct Mapping

A large part of the script performs direct one-to-one mapping from simulated source variables to export variables. This includes the main TPB variables, habit, motivational competence, action planning, self-control, the “Kompetenz-Intensions Modell” (KIM) block for intrinsic

motivation, perceived competence, perceived choice, and extrinsic motivation, as well as the core TAM items.

5.3 Value Transformations and derived variables

Several export fields require transformation rather than direct copying. Examples include conversion of 0-4 KIM responses into German-labelled Likert text, conversion of PA day counts into strings such as '1 Tag' or '3 Tage', recoding of gender and yes/no variables into survey labels.

6 Script 3: Simulation Checks and Calibration

6.1 Purpose and General Approach

The check script is used to evaluate the plausibility and internal consistency of the simulated dataset. It supports an iterative calibration process in which simulation parameters can be adjusted based on diagnostic results. Rather than serving as a formal statistical analysis, this step functions as a validation tool to ensure that the simulated data behave in a realistic and theoretically consistent manner.

6.2 Structural, Logical, and Distributional Checks

The script first verifies the overall structure of the dataset, including the distribution of participants across study groups and time points, as well as the completeness of longitudinal measurements.

It then evaluates whether the simulated data reflect the intended intervention design. Group-specific changes in physical activity and key constructs (e.g., habit) are examined to confirm that intervention effects occur at the expected time points.

In addition, logical consistency checks are performed to ensure that the data follow the survey structure. This includes verifying that:

- TAM variables are only present during relevant intervention phases,
- filter logic (e.g., days vs. minutes) is correctly implemented, and
- conditional variables behave as expected.

The script also inspects distributions of key variables to identify potential ceiling or floor effects and examines correlations between central constructs (e.g., intention, habit, planning, and physical activity) to ensure alignment with theoretical expectations.

6.3 Missing Data and Model-Based Validation

Missing data patterns are assessed at both the overall and variable-specific level, including differences across time points and key variables. This allows for evaluating whether the simulated missingness mechanisms (e.g., dropout, random missingness) behave as intended.

Finally, simple statistical models are estimated to provide an additional validation layer. Linear mixed models are used to test group $\times$ time effects, and extended models include

sociodemographic predictors. These analyses help confirm that intervention effects and covariate relationships are detectable and plausible within the simulated dataset.

7 Reproducibility and Workflow

The simulation pipeline is designed to be fully reproducible and modular, with each step implemented in a separate script.

To reproduce the full workflow, the scripts should be executed in the following order:

1. **Data simulation script:** Generates the synthetic dataset containing all variables at the participant and item level.
2. **Mapping script:** Transforms the simulated dataset into a format compatible with LimeSurvey exports.
3. **Check and calibration script:** Evaluates the resulting dataset and supports iterative refinement of simulation parameters.

The workflow requires the simulation script (R), the mapping script (R), and also a template excel file based on a LimeSurvey response export. The outputs are a CSV file containing the simulated dataset, and an Excel file that mirrors the structure of a LimeSurvey export.

All scripts rely on standard R packages (e.g., dplyr, tidyr, MASS, readxl, openxlsx, lme4) and can be executed independently once the required input files are available.

8 Key Assumptions and Design Decisions

The simulation is based on a combination of empirical evidence and pragmatic design choices. Where possible, parameters such as means, standard deviations, and correlations were derived from meta-analyses and validation studies (see appendix A1). This applies to constructs such as habit, planning, and motivational profiles (e.g., BMZI).

However, for several constructs (e.g., TPB and TAM variables), no consistent empirical benchmarks for raw score distributions were available. In these cases, parameter values were defined within plausible ranges based on theoretical expectations and typical values reported in the literature (see appendix A1).

Intervention effects were specified to reflect small effect sizes typically observed in digital physical activity interventions (e.g., Singh et al., 2024). Initial effect magnitudes were derived from empirical step-based findings and translated into weekly MET-minutes. These values were then iteratively adjusted to ensure that the simulated data showed clearly detectable but realistic group $\times$ time differences, while maintaining plausible distributions and variability across participants.

9 Limitations

Despite its theoretical grounding, the simulation has several limitations. First, the dataset is synthetic and therefore cannot fully capture the complexity and variability of real-world data. Behavioral processes are simplified, and some relationships between variables are modeled in a linear and additive way. Second, for some constructs, especially within TPB and TAM,

parameter values are based on assumptions rather than precise empirical estimates. As a result, the simulated distributions and effect sizes should be interpreted as plausible approximations rather than exact representations.

Appendix

A1 Literature-Informed Parameters Used in the Simulation

Construct	Parameter	Value	Source	Notes
<i>Habit (SRBAI)</i>	Mean (baseline)	2.11	Gardner & Lally, (2018)	Retained as baseline calibration target in the habit latent model.
	Intervention effect	0.66	Gardner & Lally, (2018)	Used as a positive treatment effect; carryover set to 25% of this value.
<i>TPB</i>	Intention–Behavior correlation	0.51	Hagger et al., (2002)	Used for correlation matrix
	PBC–Behavior correlation	0.39	Hagger et al., (2002)	Reflected conceptually in the joint modelling of PBC and PA.
<i>Planning</i>	Mean / SD	2.17 / 0.70	Pfeffer & Strobach, (2020)	Directly used
	Positive effect on PA	~0.30	Peng et al., (2022)	Slightly calibrated (adjusted downwards)
<i>Motivational Competence</i>	Mean / SD	4.38 / 0.64	Strohacker et al., (2024)	Directly used
<i>Self-control</i>	Reliability reference	$\alpha \approx 0.8\text{--}0.9$	Haktanir et al., (2024)	Used for plausibility of item consistency
<i>SDT</i>	Mean intrinsic motivation	3.4	Truong et al., (2023)	Directly used
	Mean extrinsic motivation	1.6	Truong et al., (2023)	Directly used
<i>BMZI motives</i>	Mean Fitness/Health	4.34	Schmid et al., (2018)	Directly used

	Mean Body/Appearance	3.15	Schmid et al., (2018)	Directly used
	Mean Social/Contact	2.60	Schmid et al., (2018)	Directly used
	Mean Competition	2.18	Schmid et al., (2018)	Directly used
	Mean Distraction	3.04	Schmid et al., (2018)	Directly used
	Mean Aesthetics	3.26	Schmid et al., (2018)	Directly used
	Mean Distraction/Stress reduction	3.28	Schmid et al., (2026)	Directly used
	SD Distraction/Stress reduction	1.00	Schmid et al., (2026)	Directly used
	Mean Activation/Enjoyment	3.69	Schmid et al., (2026)	Directly used
	SD Activation/Enjoyment	0.88	Schmid et al., (2026)	Directly used
	Mean Competition/Performance	2.29	Schmid et al., (2026)	Directly used
	Mean Competition/Performance	1.04	Schmid et al., (2026)	Directly used
<i>TAM</i>	Mean range	4.0–6.0	Adnan et al., (2025)	No pooled values → calibrated
<i>Physical Activity</i>	Baseline activity	410 MET min/week	Nigg et al., (2024)	Low-active sample assumption
	Intervention effect on steps (mean from studies with a 3-month long intervention period)	1842.25 steps	Edney et al., (2025)	Directly Used
	Follow-up steps effects (mean from studies with a 3-month long intervention period)	1468.33 steps	Edney et al., (2025)	Directly Used
<i>Sociodemographics</i>	Effects (age, gender, income, app use)	directional	(Pontin et al., 2021)	Directions are evidence-informed; exact sizes are calibrated.
<i>Missing data</i>	Dropout (~10% both after T1 and T2)	~20%	Loughnane et al., (2025)	Calibrated

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